

Opinion dynamics on stochastic graphs

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Abstract. This study explores how rumor transmission and opinion dynamics in social networks are shaped by polarization and individual memory capacity. To address the limitations of existing models that assume fixed memory, we propose a framework featuring tunable memory storage and dynamic affinity updates based on ideological proximity (measured via Jensen-Shannon divergence). The simulations reveal two primary findings: first, reduced memory capacity accelerates and facilitates polarization; second, within polarized groups, dissenting individuals face social exclusion without the ability to form new bonds among themselves.

Keywords. Opinion dynamics, Jensen-Shannon divergence, memory, ideological proximity.

Polarization is a widespread phenomenon that significantly affects economic, political, and legal systems [6] and often leads to the formation of echo chambers and/or homophilic clusters [1–4]. However, few existing models of rumor transmission and opinion dynamics incorporate polarization and memory capacity storage. In 2019, Wang *et al.* [7] proposed a rumor dissemination model in which information is distorted and polarized prior to transmission, with receivers deciding to accept or reject messages based on the spreader’s network connectivity. Subsequently, Rúa *et al.* [5] incorporated ideological proximity via Jensen–Shannon divergence, defining the acceptance probability as a function of both ideological alignment and spreader popularity. Notably, both frameworks evaluated their dynamics on Barabási–Albert scale-free networks while keeping memory capacity fixed.

Despite those contributions, further work is needed to account for variable memory effects on polarization. To address this gap, we propose a model in which memory capacity storage is not fixed but instead functions as a tunable parameter. To capture potential polarization effects during information propagation, we introduce a function that quantifies the polarity of information, along with a distortion probability that depends on an individual’s tendency to polarize propagated information either positively or negatively. This approach offers greater realism in modeling information diffusion: rather than starting from a blank slate and filling their memories solely with others’ opinions, individuals transmit information from their own existing, biased, and variable memories.

We also presents a model that considers the affinity between individuals on a stochastic graph. The affinity ingredient is an innovative feature that simulates the updating of neighbors (edges) in the graph. It is defined through the Jensen-Shannon measure, which quantifies the ideological proximity between individuals and is identified as the weight of network connections.

The main conclusions of our work are twofold. First, a reduced memory capacity facilitates the polarization process. Second, within polarized groups, individuals who disagree with the majority are excluded and fail to form new bonds with other excluded members, preventing the emergence of an alternative group.

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